OpenGeoSys Training 2026 – KIGAM

Lecture 3: Hydromechanics - Groundwater Flow

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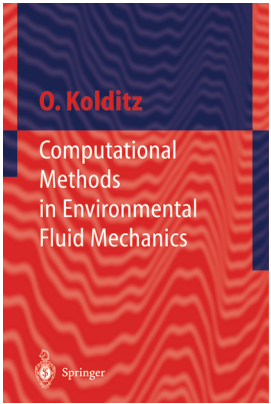
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Daejeon, 11th October 2026

DECOVALEX 2027

Agenda

1. OGS Course Overview
2. Hydromechanics
   - OGS-H
   - General Balance Equation
   - (Phase) Mass Conservation
   - Divergence-Free Flow
   - Groundwater flow: Hydraulic head formulation
   - Groundwater flow: Pressure formulation
3. References

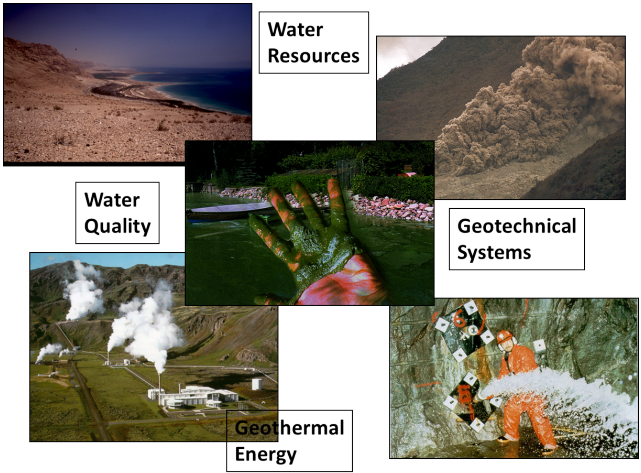


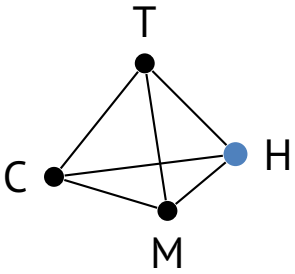
Lecture 2

Hydromechanics

Groundwater Flow

OpenGeoSys H





Starting point: General Balance Equation:

$$\frac{d\psi}{dt} = \frac{\partial \psi}{\partial t} + \nabla \cdot (\mathbf{v}\psi) - \nabla \cdot (\mathbf{D}^{\psi} \nabla \psi) = Q^{\psi} \tag{1}$$

The differential equation of mass conservation ($\psi = \rho$) in divergence form becomes

$$\frac{d\rho}{dt} = \frac{\partial \rho}{\partial t} + \nabla \cdot (\mathbf{v}\rho) = 0 \tag{2}$$

There is no diffusion in phase mass conservation: $\mathbf{D}^{\rho} = 0$

Partial differentiation of the above equation gives

$$\frac{d\rho}{dt} = \frac{\partial \rho}{\partial t} + \mathbf{v} \cdot \nabla \rho + \rho \nabla \cdot \mathbf{v} = 0 \tag{3}$$

Using the material (or convective) derivative the mass conservation equation can be rewritten as

$$\frac{\partial \rho}{\partial t} = -\rho \nabla \cdot \mathbf{v} \tag{4}$$

Note, above convective form of mass conservation equation becomes zero only for incompressible flows, i.e.

$$\frac{\partial \rho}{\partial t} = 0 \tag{5}$$

requires divergence-free flow.

$$\nabla \cdot \mathbf{v} = 0 \tag{6}$$

From equation (3) results that the above expression is the continuity equation for a homogeneous fluid ($\rho = \text{const}$).

$$\nabla \cdot \mathbf{v} = 0 \tag{7}$$

which in 2D becomes

$$\frac{\partial v_y}{\partial x} + \frac{\partial v_y}{\partial y} = 0 \tag{8}$$

Links:

- https://de.wikipedia.org/wiki/Stromfunktion
- https://www.ingenieurkurse.de/stroemungslehre/ebene-stroemungen/quelle-und-senke-divergenz.html

Solutions of equation (8) are

$$v_x = \frac{x-L}{(x-L)^2+y^2} - \frac{x+L}{(x+L)^2+y^2} \tag{9}$$

$$v_y = \frac{y}{(x-L)^2+y^2} - \frac{y}{(x+L)^2+y^2} \tag{10}$$

After deriving v_X and v_y, respectively, we can proof equation (8)

$$\frac{\partial v_x}{\partial x} = \frac{y^2-(x-L)^2}{((x-L)^2+y^2)^2} - \frac{y^2-(x+L)^2}{((x+L)^2+y^2)^2} \tag{11}$$

$$\frac{\partial v_y}{\partial y} = \frac{(x-L)^2-y^2}{((x-L)^2+y^2)^2} - \frac{(x+L)^2-y^2}{((x+L)^2+y^2)^2} \tag{12}$$

Exercise
Divergence free flow

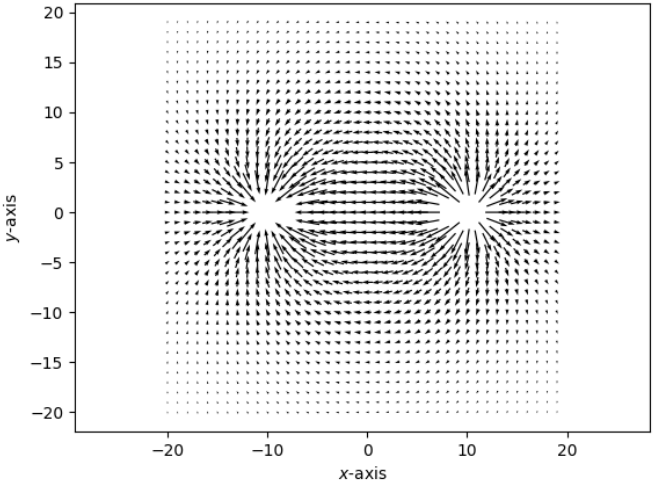
```python
import numpy as np
import matplotlib.pyplot as plt
# set up a normalized grid:
dim= 20
xarray= np.arange(-dim,dim)
yarray= np.arange(-dim,dim)
# (fluid) flow from a source at L to a sink at -L:
L = dim/2
x,y = np.meshgrid(xarray,yarray)
vx = (x-L)/((x-L)**2+y**2) - (x+L)/((x+L)**2 +y**2)
vy = y/((x-L)**2+y**2) - y/((x+L)**2 +y**2)
# plot the flow lines:
plt.figure()
plt.quiver(x,y, vx, vy, pivot='mid')
plt.xlabel("$x$-axis")
plt.ylabel("$y$-axis")
plt.axis('equal')
plt.show()
```

Listing: Python code for divergence-free flow - div v

Source: University of Auckland

Hydromechanics

Divergence free flow: EX22-divergenzfreie-stroemung.py



Repo: https://github.com/OlafKolditz/Hydroinformatik-2026/blob/main/EX22-divergenzfreie-stroemung.py

Source: https://auckland.figshare.com/articles/dataset/Chapter_6_Divergence_of_a_vector_field/5732421

Lecture
Groundwater flow equation

Mass balance: fluid in pore space

$$\frac{\partial n_e \rho}{\partial t} + \nabla \cdot (n_e \rho \mathbf{v}) = Q_\rho \tag{13}$$

Incompressible fluid (ρ_w = constant)

$$\rho_w \frac{\partial n_e}{\partial t} + \rho_w \nabla \cdot (n_e \mathbf{v}) = Q_\rho \tag{14}$$

$$\frac{\partial n_e}{\partial t} + \nabla \cdot (n_e \mathbf{v}) = \frac{Q_\rho}{\rho_0} \tag{15}$$

Groundwater hydraulics

$$\frac{\partial n_e}{\partial t} = S_0 \frac{\partial h}{\partial t} \tag{16}$$

$$n_e \mathbf{v} = \mathbf{q} = -\mathbf{K} \nabla h \quad \text{(Darcy' law)}$$

$$
\begin{aligned}
S_0 \frac{\partial h}{\partial t} - \nabla \cdot (\mathbf{K} \nabla h) &= Q \\
S_0 \frac{\partial h}{\partial t} - \frac{\partial}{\partial x}\left(K_x \frac{\partial h}{\partial x}\right) - \frac{\partial}{\partial y}\left(K_y \frac{\partial h}{\partial y}\right) - \frac{\partial}{\partial z}\left(K_z \frac{\partial h}{\partial z}\right) &= Q \qquad (17)
\end{aligned}
$$

For 2D horizontal groundwater flow we have

$$
S_0 \frac{\partial h}{\partial t} - \frac{\partial}{\partial x}\left(K_x \frac{\partial h}{\partial x}\right) - \frac{\partial}{\partial y}\left(K_y \frac{\partial h}{\partial y}\right) = Q \qquad (18)
$$

Mass balance: fluid in pore space

$$\frac{\partial n_e \rho}{\partial t} + \nabla \cdot (n_e \rho \mathbf{v}) = Q_\rho \tag{19}$$

Mass balance formulation

$$\left(\rho_w \frac{\partial n_e}{\partial p_w} + n_e \frac{\partial \rho_w}{\partial p_w} \right) \frac{\partial p_w}{\partial t} - \nabla \cdot \left(\rho_w \frac{\mathbf{K_0}}{\mu_w} (\nabla p_w - \rho_w \mathbf{g}) \right) = \rho_w W_0 \tag{20}$$

Volume balance formulation (ρ_w = constant)

$$\frac{\partial n_e}{\partial p_w} \frac{\partial p_w}{\partial t} - \nabla \cdot \left(\frac{\mathbf{K_0}}{\mu_w} (\nabla p_w - \rho_w \mathbf{g}) \right) = W_0 \quad , \quad \beta_S = \partial n_e / \partial p_w \tag{21}$$

$$\text{Darcy velocity:} \quad \mathbf{q} = -\frac{\mathbf{K_0}}{\mu_w} (\nabla p_w - \rho_w \mathbf{g}) \tag{22}$$

Groundwater flow equation

Parameters

Symbol	Unit	SI Unit	Definition
S_0	[1/L]	[1/m]	specific storage coefficient in hydrogeology
ρ_w , (ρ^f)	[M/L^3]	[kg/m^3]	fluid density
g	[L/T^2]	[m/s^2]	gravitational acceleration coefficient
$\beta_s = \partial n_e / \partial p_w$	[LT^2/M]	[1/Pa]	storage coefficient of OGS
p_w , (p^f)	[M/LT^2]	[Pa]	liquid phase (i.e. water) pressure
t	[T]	[s]	time
∇	[1/L]	[1/m]	divergence, gradient operator
$\mathbf{K_0}$	[L^2]	[m^2]	intrinsic permeability (tensor)
$\mathbf{K}$	[L/T]	[m/s]	hydraulic conductivity (tensor)
$\mathbf{g}$	[L/T^2]	[m/s^2]	gravitational acceleration vector (0,0,g)
μ_w	[M/LT]	[Pa s]	dynamic fluid viscosity
n_e	[1/T]	[1/s]	effective porosity
n_e	[-]	[-]	effective porosity
h	[L]	[m]	hydraulic head
W_0	[1/T]	[1/s]	source/sink term

References

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